

Tuning Direct CO Hydrogenation Reaction over Fe-Mn Bimetallic Catalysts toward Light Olefins: Effects of Mn Promotion

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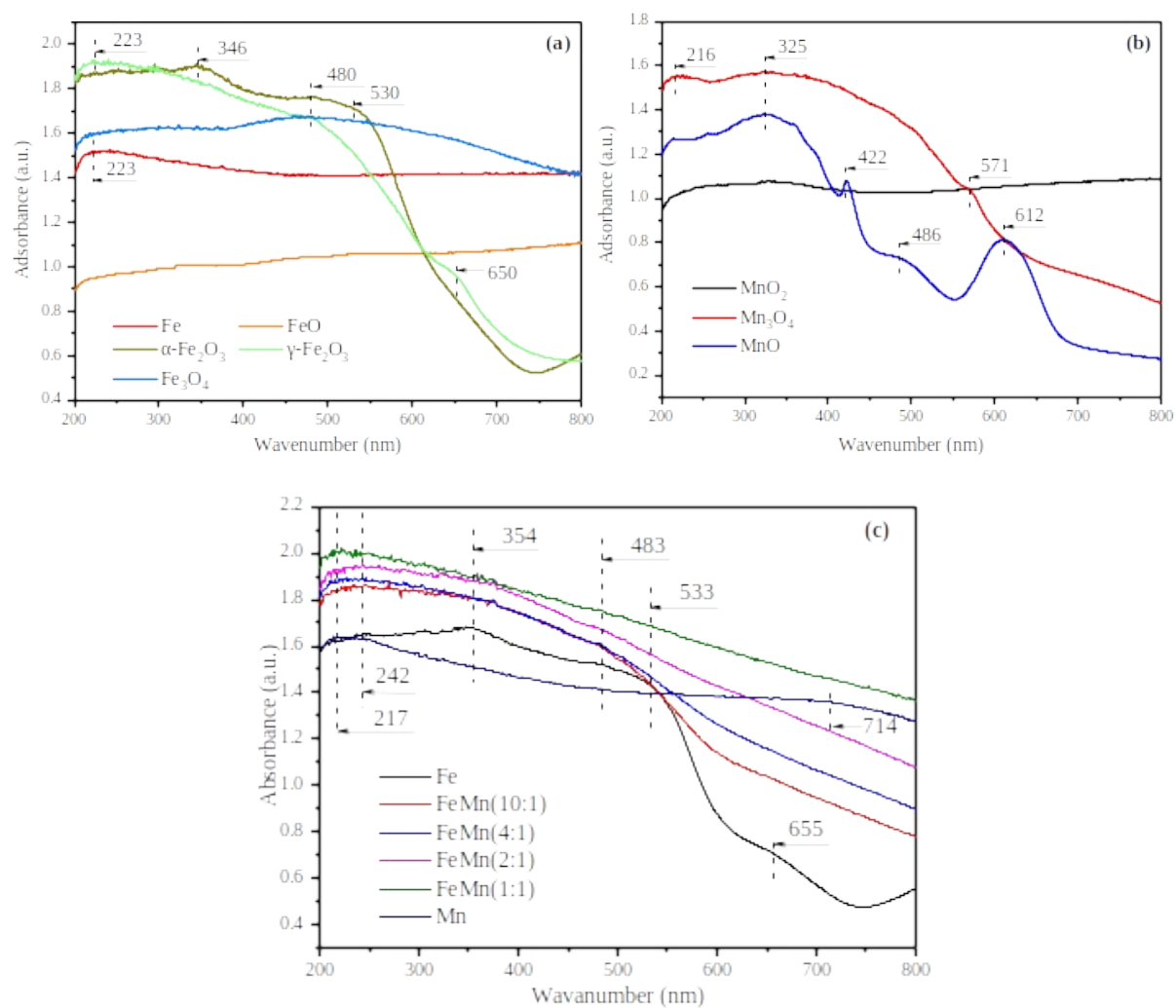


Fig. S1. Diffuse reflectance UV-Vis spectra of Fe/FeO_x standards (a), MnO_x standards (b), and the as-prepared Fe-Mn bimetallic catalysts (c).

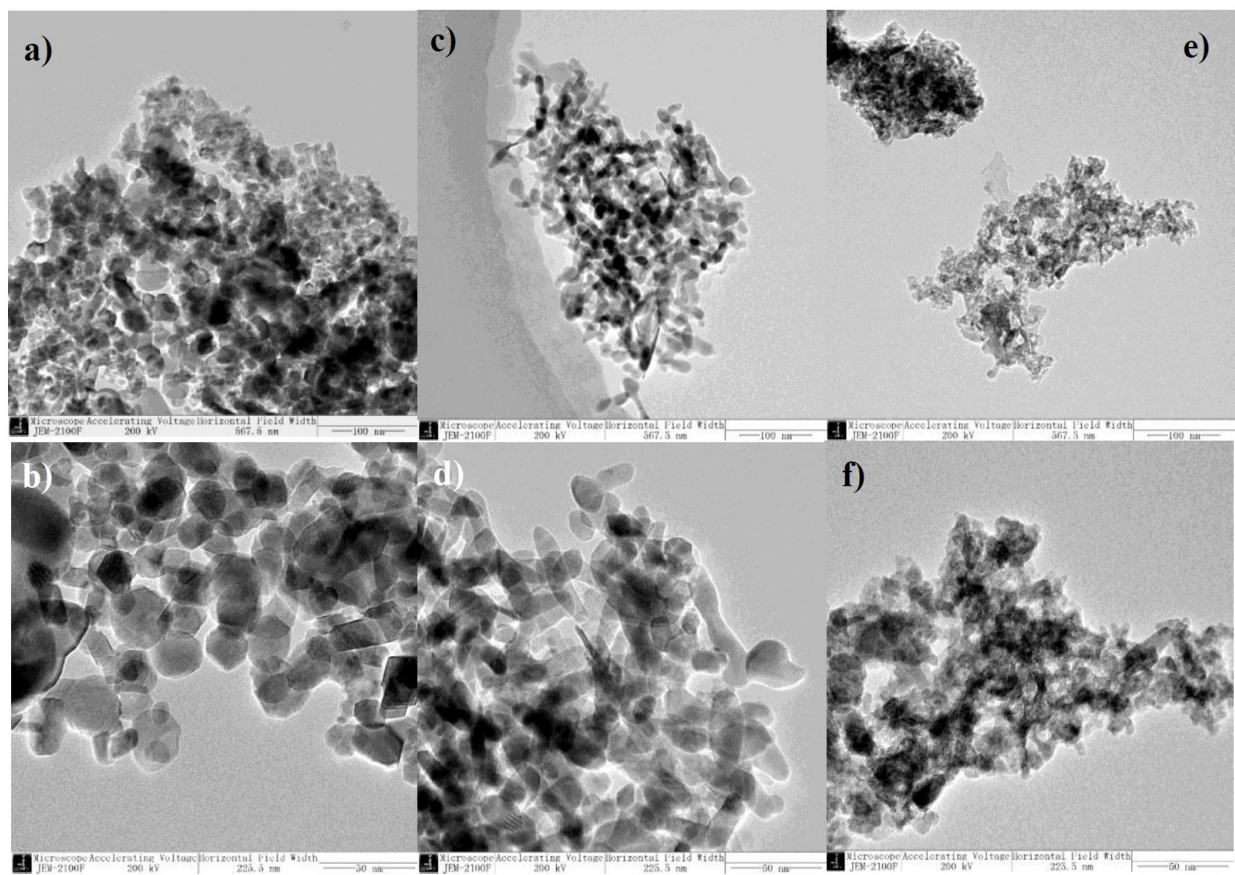


Fig. S2. TEM images of the as-prepared Fe (a, b), FeMn(4:1) (c, d), FeMn(1:1) (e, f) catalysts

Table S1. CO conversion and product selectivity of FTS over different Fe-Mn bimetallic catalysts. Reduction conditions: H₂ (99.999 vol. %), 25 ml/min, 350 °C, 0.1 MPa, 5 h. Reaction conditions: catalyst diluted with SiC (mixing ratio=1:3), 260 °C, 2.0 MPa, H₂/CO=1, gas flow=25ml/min, TOS=20 h.

Catalysts	Conv.% (20 h)	Carbon Selectivity (%)								O/P ratio		α	FTY ($\mu\text{mol}_{\text{CO}} \cdot g_{\text{Fe}}^{-1} \cdot \text{s}$)	C ₂ -C ₄ = selectivity, CO ₂ free
		CH ₄	C ₂ =	C ₃ =	C ₄ =	C ₂ -C ₄ =	C ₅ +	CO ₂	Oxy	C ₂	C ₂ -C ₇			
Pure Fe	2.62	11.76	8.29	11.82	6.00	26.11	10.93	44.10	1.35	3.17	4.04	0.67	9.75	46.7
FeMn(10:1)	4.62	14.60	10.06	16.26	8.73	35.05	10.76	27.64	2.05	2.12	3.11	0.53	18.9	48.4
FeMn(4:1)	5.49	15.02	9.55	18.95	10.13	38.63	10.48	20.72	2.42	1.55	2.89	0.48	25.5	48.7
FeMn(2:1)	3.52	11.36	10.79	17.13	8.16	36.08	15.33	24.80	3.02	2.98	3.66	0.71	19.5	47.9
FeMn(1:1)	1.98	11.81	10.06	15.59	7.83	33.48	15.86	26.47	3.15	2.96	3.44	0.72	14.7	45.5

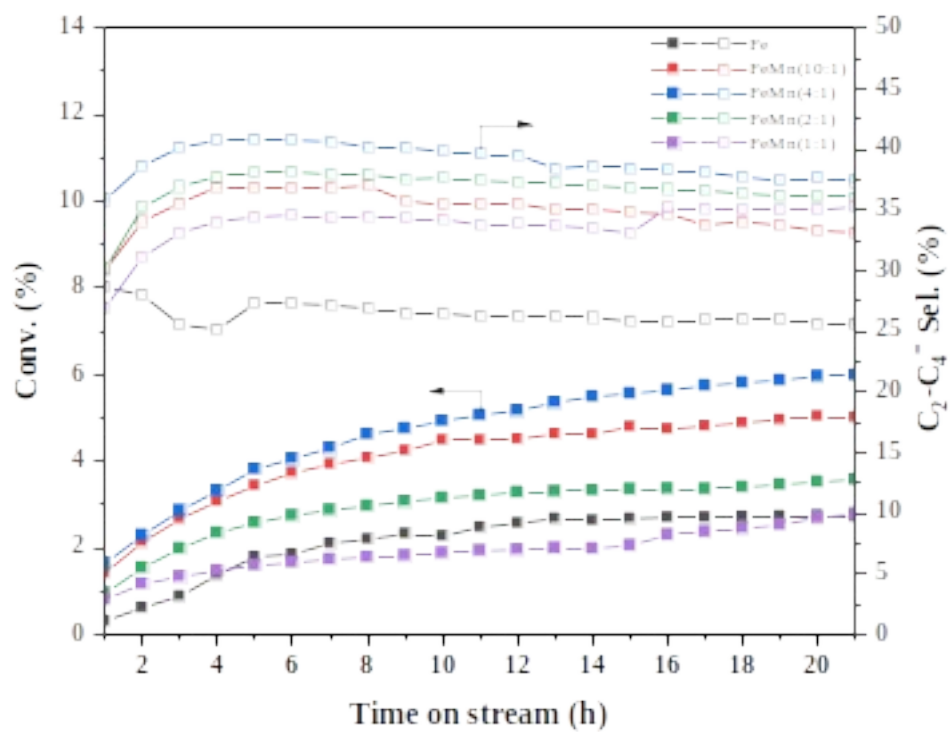


Fig. S3. Catalytic performance over Fe-Mn bimetallic catalysts with 20h time on stream. C2-C4 olefin selectivity is calculated on a carbon basis including CO₂.

Table S2. Brief comparison of catalytic performance over the FeMn catalysts for Fischer-Tropsch to olefins

FeMn catalysts	Preparation Method ^a	CO Conv %	FTY, $\mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{Fe}} \cdot \text{s}^{-1}$	CO ₂ Sel %	Product distributions, %			O/P ratio ^b
					CH ₄	C ₂ -C ₄	C ₅ +	
FeMnCuSiO ₂ (100/10/3/12) [1] ^c	Prec.+Imp.	96.9	5.3 ^d	23.0	20.0	40.1	9.5	1.32
FeMn(3/1)-2wt%K[2] ^e	Prec.+Imp.	97.65	-	29.86	14.5 9	26.3 6	53.7 8	5.00
100Fe/33Mn/2.8Na[3, 4] ^f	Prec.+Imp.	96.2	-	26.8	16.7	30.6	44.3	3.6
Fe ₂₀ -Mn _{1.0} /SiO ₂ [5] ^g	Imp.	14.4	-	23.3	21.6	40.3	19.4	2.6
Al-Fiber@meso-Al ₂ O ₃ @Fe-Mn-K[6] ^h	Imp.	90.0	126.3	31.1	15.7	38.8	35.5	3.9
Fe _{2.86} Mn _{0.14} O ₄ /CNT[7] ⁱ	Thermo.+ULA	43.9	275	37.2	6.1	31.5	53.8	3.62
Fe/MnK-CNTs[8] ^j	Redox.+Prec.	31.8	261.2	36.9	32.4	50.4	9.1	6.2
Mn-07[9] ^k	Prec.	45.1	-	19.2	12.0	33.2	44.2	3.17
0.5Mn/ γ -Fe ₂ O ₃ [10] ^l	Prec.+Imp.	55.1	14.8	31.7	11.7	61.2	18.2	6.9
6%Mn/Fe ₃ O ₄ [11] ^m	Solv.+Imp.	41.5	9.9	37.8	9.7	60.1	23.7	9.2
Fe-13Mn[12] ⁿ	Solv.	7.1	16.9	25.2	17.0	54.4	13.3	3.6
FeMn(4:1) ^o	Prec.	5.5	25.5	20.7	18.9	48.7	16.1	3.04

^a Methods: Imp.=impregnation, Prec.=precipitation, Solv.=Solvothermal, Redox.=reduction-oxidation, Themo.=thermal decomposition, ULA= ultrasonic assembly

^b C₂-C₄ olefins/paraffins

^c Reaction conditions: 573K, H₂/CO=2.0, 2.0MPa, 1500h⁻¹

^d Estimated from FTY=3.09 $\mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{cat}} \cdot \text{s}^{-1}$

^e Reaction conditions: 593K, H₂/CO=2.0, 1.5MPa, 1000h⁻¹

^f Reaction conditions: 598K, H₂/CO=3.5, 1.0MPa, 2000h⁻¹, 36h

^g Reaction conditions: 533K, H₂/CO=1.0, 2.0MPa, 4000h⁻¹

^h Reaction conditions: 623K, H₂/CO=2.0, 4.0MPa, 6000ml/(h g_{cat})

ⁱ Reaction conditions: 573K, H₂/CO=1.0, 2.0MPa, 6000ml/(h g_{cat}), 12h

^j Reaction conditions: 543K, H₂/CO=1.0, 2.0MPa, 24000ml/(h g_{cat})

^k Reaction conditions: 523K, H₂/CO=2.0, 1.5MPa, 2000h⁻¹

^l Reaction conditions: 523K, H₂/CO=2.0, 1.5MPa, 2000h⁻¹, 20h

^m Reaction conditions: 593K, H₂/CO=1.0, 1.0MPa, 0.1mol/(h g_{cat})

ⁿ Reaction conditions: 533K, H₂/CO=1.0, 2.0MPa, 4000h⁻¹

^o Reaction conditions: 533K, H₂/CO=1.0, 2.0MPa, 15000ml/(h g_{cat}) (This work)

Table S3. EXAFS fitting parameters at the Fe K-edge for various samples ^a.

Sample	Shell	CN	R(Å)	σ^2 (Å ²)	ΔE_0 (eV)	R factor
Fe foil	Fe-Fe (1 st)	8	2.46±0.01			
	Fe-Fe (2 nd)	6	2.84±0.01	0.0049	5.2±1.0	0.0051
Spent Fe	Fe-C	3.6±0.5	1.99±0.02	0.0087		
	Fe-Fe	8.4±0.4	2.59±0.01	0.0139	-6.1±1.5	0.0114
Spent FeMn(4:1)	Fe-C	1.9±0.9	1.81±0.06	0.0032		
	Fe-O	2.9±0.8	1.97±0.05	0.0053		
	Fe-Fe (1 st)	4.1±0.5	2.59±0.03		-4.5±4.7	0.0140
	Fe-Fe (2 nd)	2.8±0.8	3.11±0.05	0.0078		

^aCN: coordination numbers; R: bond distance; σ^2 : Debye-Waller factors; ΔE_0 : the inner potential correction. R factor: goodness of fit. S_0^2 was set to 0.71, according to the experimental EXAFS fit of Fe foil reference by fixing CN as the known crystallographic value.

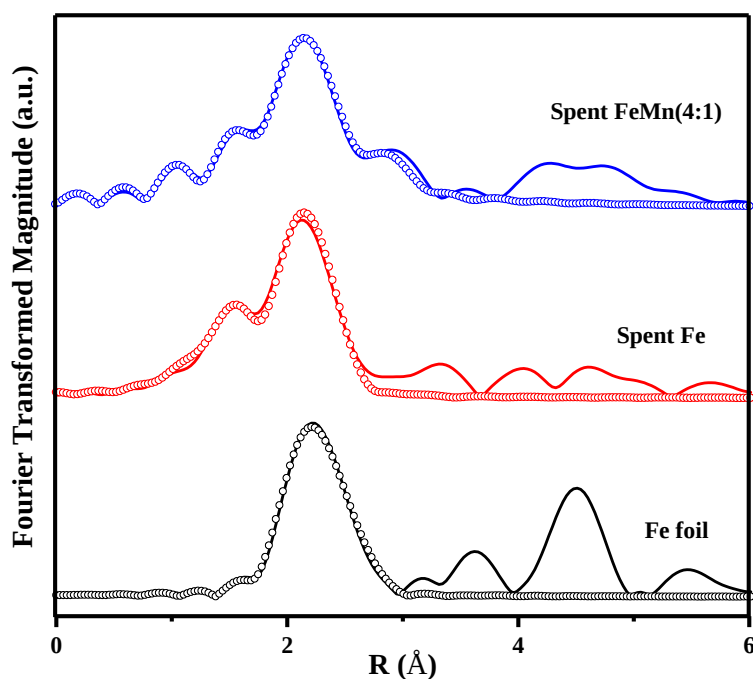


Fig. S4. Fourier transformed Fe K-edge EXAFS data of the spent FeMn catalysts and the Fe foil standard. The lines represent experimental data, and the circles are fitted data.

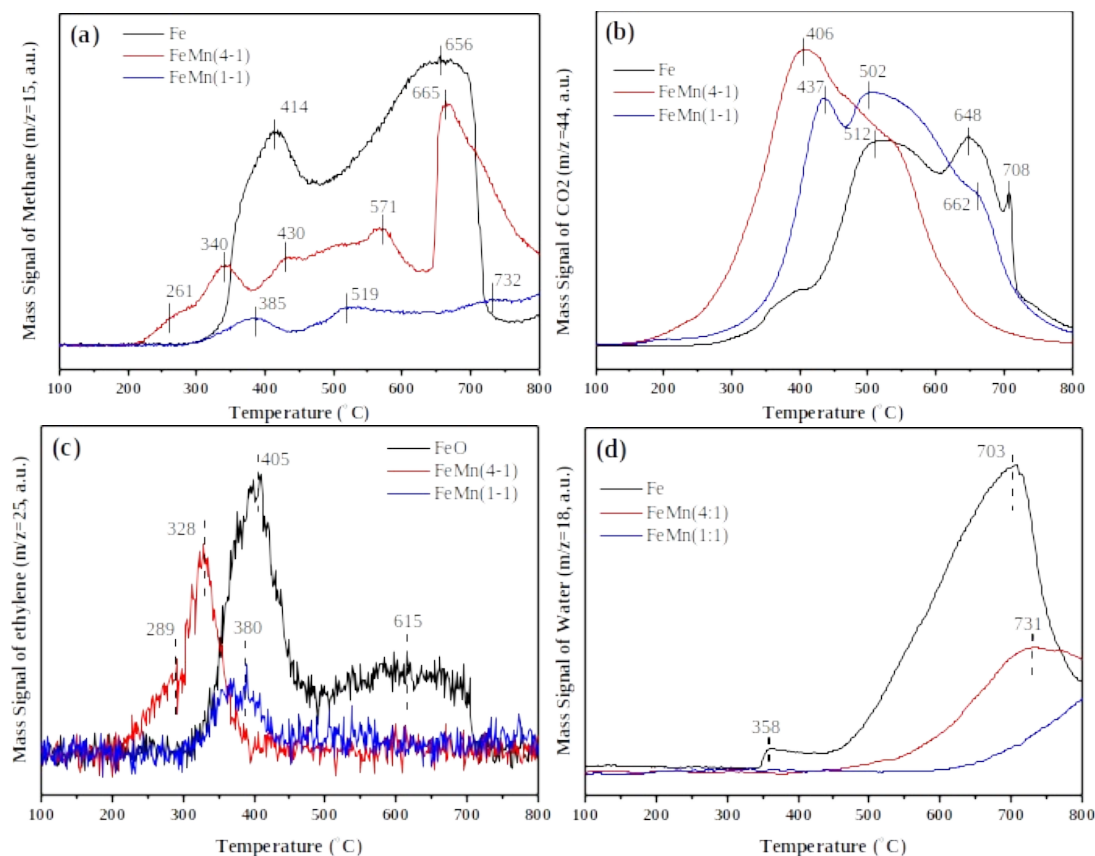


Fig. S5. CO/H₂-TPSR profiles of H₂-reduced catalysts. (a) CH₄, (b) CO₂, (c) C₂H₄, (d) H₂O, Fe, FeMn(4:1), and FeMn(1:1) are denoted by black, red, and blue solid lines, respectively.

Syngas temperature programmed surface reaction (CO/H₂-TPSR) was performed to investigate the surface activities of the H₂-reduced catalysts. Desorption profiles of CH₄ (m/z=15), CO₂ (m/z=44), C₂H₄ (m/z=25), and H₂O (m/z=18) are shown in Fig. S3. CH₄ desorption occurs in two major temperature regions, where the lower temperature (300-500 °C) desorption is attributed to the hydrogenation of the iron carbides, and the high temperature (>600 °C) is related to the hydrogenation of coke deposition on catalyst surface [13]. The first desorption peak locates at 414, 261 and 385 °C for Fe, FeMn(4:1) and FeMn(1:1), respectively, suggesting that CH₄ is more easily desorbed from the Mn-containing catalyst surface. The flat desorption profile of FeMn(1:1) suggests that excess Mn restrains the transformation of iron oxides to iron carbides, and therefore disfavors the formation of CH₄. Direct CO dissociation gives chemisorbed oxygen

atoms that desorb as CO_2 . The CO_2 desorption profiles show that the first desorption peak appears at 406, 437, and 512 °C for FeMn(4:1), FeMn(1:1) and Fe catalysts, respectively, further confirming that the addition of Mn promotes the direct CO dissociation ability of the catalysts. Moreover, the lower C_2H_4 desorption temperature on Fe-Mn bimetallic catalyst surface indicates that the addition of Mn favors the desorption of olefins. An intense water desorption peak appears in the region of 700-800 °C, corresponding to the removal of lattice oxygen in the oxides. The water desorption temperature shows an increasing tendency as the Mn content increases, indicating that high Mn level increases the stability of the Fe/Mn oxides. These results confirm that the surface activity of the Fe catalyst can be enhanced by adding appropriate amount of Mn. It is reported that appropriate amount of Mn can improve the surface basicity of the catalysts, which exhibit excellent CO dissociation ability while suppressing the hydrogenation ability [14].

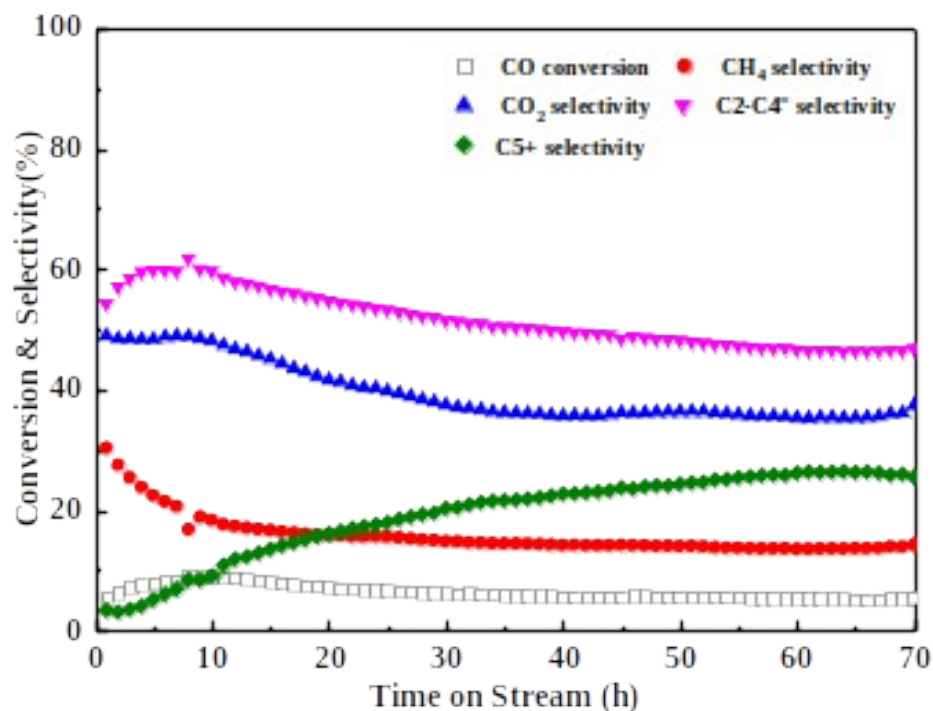


Fig. S6. Stability test of FeMn(4:1) catalyst. The catalyst is pelleted, crushed, and sieved, physically mixed with SiO₂ in 40-60 mesh. Activation Conditions: 5%CO/5%H₂/Ar, 100mL/min, 2°C/min, T=350°C, 5h. Reaction Conditions: T=260°C, P=2.0 MPa, Syngas(H₂/CO/N₂=45/45/10), GHSV=21600 h⁻¹. The selectivities of hydrocarbons are calculated on carbon mole percent excluding CO₂ in the product stream.

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